Capstone Project

the World Bank World Development Indicators Dataset

Predicting Carbon Dioxide (CO_2) Emissions Using Socioeconomic and Environmental Indicators

Submitted by

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1.0 Introduction

Climate change is one of the world's most pressing environmental challenges. Carbon dioxide (CO_2) emissions contribute significantly to global warming and are influenced by socioeconomic factors such as energy use, economic growth, urbanization and access to electricity. Predicting CO_2 emissions using development indicators can support policymakers in designing sustainable environmental policies.

This project applies machine learning techniques to predict CO_2 emissions using the World Bank World Development Indicators (WDI) dataset.

2.0 Problem Statement

Governments seek to reduce carbon emissions while supporting economic development. However, increasing energy consumption, urbanization and economic growth often lead to higher emissions. This project investigates whether selected socioeconomic and environmental indicators can accurately predict national CO_2 emissions.

3.0 Methodology

The World Bank World Development Indicators dataset (2023) was used for this analysis. Data preprocessing involved selecting relevant indicators, removing regional aggregates, handling missing values using median imputation and preparing the dataset for machine learning.

The selected predictors were:

1. Access to electricity
2. Energy use
3. Forest area
4. GDP per capita
5. Urban population

The target variable was:

CO_2 emissions excluding LULUCF per capita

Two regression models were developed:

a. Linear Regression
b. Random Forest Regressor

4.0 Exploratory Data Analysis (EDA)

Key findings include:

1. Energy use showed the strongest positive relationship with CO_2 emissions;
2. GDP per capita had a moderate positive relationship;
3. Urban population also showed a positive association;
4. Forest area had a relatively weak relationship.
5. Correlation analysis and visualizations were used to understand these relationships before model development.

5.0 Model Evaluation

Model	**MAE**	**RMSE**	**R^2**
Linear Regression	1.69	2.17	0.52
Random Forest	2.71	7.43	-4.64

Linear Regression performed better than Random Forest and was selected as the final model.

6.0 Recommendations

1. Increase investment in renewable energy;
2. Promote sustainable urban development;
3. Improve energy efficiency;
4. Strengthen environmental data collection and reporting.

7.0 Conclusion

This project demonstrated that socioeconomic and environmental indicators can be used to predict national CO_2 emissions. Linear Regression achieved the best performance, explaining approximately 52% of the variation in CO_2 emissions. The findings highlight the importance of energy use, economic development and urbanization in shaping carbon emissions and can support evidence-based environmental decision-making.